

UnderDeterminedProblems

Consider $y = Ax$ where $y \in \mathbb{R}^p$, $x \in \mathbb{R}^n$, $n > p$. The equations are consistent if $\exists y \in \mathbb{R}^p, \exists x_0 \in \mathbb{R}^n$ s.t. $y = Ax_0$. We define the set of solutions

$$V = \{x \in \mathbb{R}^n | y = Ax\} \neq \emptyset$$

if consistent. If we define $\mathcal{N}(A) = \{x \in \mathbb{R}^n | Ax = 0\}$ = null space of A , then clearly, if x_0 is one solution of the equations, then $\forall \bar{x} \in \mathcal{N}(A)$, $x_0 + \bar{x}$ is also a solution, that is

$$y = A(x_0 + \bar{x})$$

Exercise (Easy) If $y = Ax_0$, then $V = \{x \in \mathbb{R}^n | y = Ax\} = x_0 + \mathcal{N}(A)$ □

How to choose among the set of solutions?

Common choice: take the one of minimum norm

$$\hat{x} = \arg \min_{x \in V} \|x\|.$$

Here is the general version of the problem. Let $(\mathcal{X}, \mathcal{F}, \langle \cdot, \cdot \rangle)$ be an inner product space, let $\{y_1, \dots, y_p\}$ be a linearly independent set, and let $c_1, \dots, c_p \in \mathcal{F}$ be a set of scalars. Define $V = \{x \in \mathcal{X} | \langle x, y_i \rangle = c_i, i = 1, \dots, p\}$

Exercise: $\exists! x_0 \in \text{span}\{y_1, \dots, y_p\}$ such that $\langle x_0, y_i \rangle = c_i, i = 1, \dots, p$.

Remark: Equivalent: $\exists! x_0 \in \mathcal{X}$ s.t. $V \cap \text{span}\{y_1, \dots, y_p\} = \{x_0\}$ □

Solution: Write $x_0 = \beta_1 y_1 + \dots + \beta_p y_p$, obtain the normal equations, and deduce that the Gram matrix is nonsingular. □

Notation: $[y_1, \dots, y_p] := \text{span}\{y_1, \dots, y_p\}$.

Proposition $V = x_0 + [y_1, \dots, y_p]^\perp$

Proof

To show: $x \in V \iff x - x_0 \perp [y_1, \dots, y_p]$

$$x \in V \iff \langle x, y_i \rangle = c_i, i = 1, \dots, p$$

$$\iff \langle x - x_0, y_i \rangle = \langle x, y_i \rangle - \langle x_0, y_i \rangle = c_i - c_i = 0, i = 1, \dots, p$$

$$\iff (x - x_0) \perp [y_1, \dots, y_p]$$

□

Theorem Let $(\mathcal{X}, \mathcal{F}, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space and $\{y_1, \dots, y_p\}$ a linearly independent set, and $c_1, \dots, c_m$ a set of scalars. Then $\exists! v^* \in V = \{x \in \mathcal{X} | \langle x, y_i \rangle = c_i, i = 1, \dots, p\}$ having minimum norm, and v^* is characterized by $v^* \perp [y_1, \dots, y_p]^\perp$ (not $v^* \perp V$).

Proof: Let $M = [y_1, \dots, y_p]^\perp$ and write $V = x_0 + M$. The problem is to determine

$$\inf_{v \in V} \|v\| = \inf_{m \in M} \|x_0 + m\| = \inf_{m \in M} \|x_0 - m\|$$

because M is a subspace. By the Projection Theorem, $\exists! x^* \in M$ such that

$$\|x_0 - x^*\| = \inf_{m \in M} \|x_0 - m\|,$$

and x^* is characterized by $x_0 - x^* \perp M$. We set $v^* = x_0 - x^*$, and then

$$\|v^*\| = \inf_{v \in V} \|v\| \quad \text{and} \quad v^* \perp M.$$

□

Remark $v^* \perp [y_1, \dots, y_p]^\perp \iff v^* \in ([y_1, \dots, y_p]^\perp)^\perp \iff v^* \in [y_1, \dots, y_p]$

Because $\mathcal{X} = [y_1, \dots, y_p] \oplus [y_1, \dots, y_p]^\perp$

△

Theorem Let $(\mathcal{X}, \mathcal{F}, \langle \cdot, \cdot \rangle)$ be an inner product space, $\{y_1, \dots, y_p\}$ a linearly independent set, and $c_1, \dots, c_p$ constants. Let

$$V = \{x \in \mathcal{X} \mid \langle x, y_i \rangle = c_i, i = 1, \dots, p\}$$

Then

$$\exists! v^* \in V \text{ s.t. } \|v^*\| = \inf_{v \in V} \|v\|.$$

Moreover, $v^* = \sum_{i=1}^p \beta_i y_i$

Where the β_i 's satisfy the normal equations:

$$\begin{bmatrix} \langle y_1, y_1 \rangle & \langle y_2, y_1 \rangle & \cdots & \langle y_p, y_1 \rangle \\ \vdots & \vdots & & \vdots \\ \langle y_1, y_p \rangle & \langle y_2, y_p \rangle & \cdots & \langle y_p, y_p \rangle \end{bmatrix} \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$$

Proof From our remark, $v^* \in [y_1, \dots, y_p]$. We write $v^* = \beta_1 y_1 + \dots + \beta_p y_p$ and then impose $v^* \in V$. This gives the normal equations. □

Return to $y = Ax$

$y \in \mathbb{R}^p, x \in \mathbb{R}^n, n > p, y = Ax.$

We define an inner product on $\mathbb{R}^n$ by $\langle x, \tilde{x} \rangle = x^\top Q \tilde{x}$, $Q > 0$. So that $\|x\| = (x^\top Q x)^{1/2}$.

Theorem If the rows of A are linearly independent, then

$$\hat{x} = \arg \min_{Ax=b} \|x\|$$

is given by

$$\boxed{\hat{x} = Q^{-1} A^\top (A Q^{-1} A^\top)^{-1} b}$$

Proof We write $A = \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix}$ and $\tilde{A} = A Q^{-1}$, define $v_i \in \mathbb{R}^n$ by $v_i = (a_i Q^{-1})^\top = Q^{-1} a_i^\top$ because Q is symmetric. It follows that

$$Ax = b \iff \langle v_i, x \rangle = b_i, \quad i = 1, \dots, p$$

Hence,

$$\hat{x} = x^* = \sum_{i=1}^p \beta_i v_i = [v_1 | \dots | v_p] \beta$$

and $G\beta = b$ and G is the Gram matrix.

Exercise

$$G = AQ^{-1}A^\top \quad \text{and} \quad G = G^\top$$

$$[v_1 | \dots | v_p] = Q^{-1}A^\top$$

□

Hence,

$$\hat{x} = Q^{-1}A^\top(AQ^{-1}A^\top)^{-1}b$$

□

Solution to Exercise

$$v_i = Q^{-1}a_i^\top$$

$$[v_1 | \dots | v_p] = [Q^{-1}a_1^\top | \dots | Q^{-1}a_p^\top]$$

$$= Q^{-1}[a_1^\top | \dots | a_p^\top]$$

$$= Q^{-1}A^\top$$

$$G_{ij} = \langle v_i, v_j \rangle$$

$$= v_i^\top Q v_j$$

$$= (a_i Q^{-1}) Q Q^{-1} a_j^\top$$

$$= a_i Q^{-1} a_j^\top$$

$$(AQ^{-1}A^\top)_{ij} = a_i (Q^{-1}A^\top)_{j\text{-column}}$$

$$= a_i Q^{-1} a_j^\top$$

□